

Taslim R. Anupom, Ph.D.

CONTACT INFORMATION	303 E 34th St. Wolfforth Texas 79382 USA	Voice: (713) 355-7542 E-mail: anupom458@gmail.com LinkedIn: www.linkedin.com/in/anupom458/
SKILLS	Electrical Engineer, Microfluidics, Systems Design, UX Design, Inventions, Space Research, Project & Team Management	
EDUCATION	Texas Tech University , Lubbock, Texas USA Ph.D., Electrical Engineering, May 2022 - Dissertation Topic: “Development of Integrated Microfluidic Platforms for Phenotyping Studies in the Model Organism <i>C. elegans</i>” - Advisor: Siva A. Vanapalli Texas Tech University , Lubbock, Texas USA M.S., Electrical Engineering, May 2020 Texas Tech University , Lubbock, Texas USA B.S., Electrical Engineering, December 2017	
HONORS AND AWARDS	American Society for Gravitational and Space Research(ASGSR), 2018 TechConnect World Innovation Conference and Expo, 2018	
EXPERIENCES	NemaLife Inc., Lubbock, Texas, USA <i>Lead Platform Developer</i> January 2023 - present - Core Platform Architecture Development: Designed and implemented a scalable platform architecture to support the organization’s present and future needs - Team Leadership: Managed and mentored six R&D team members to ensure the delivery of high-quality products aligned with organizational standards and best practices. Also managed eight data analysis team members to efficiently meet data delivery timeline to clients and internal asset generation projects - Cross-Functional Collaboration: Worked closely with stakeholders and departments to ensure the platform met business objectives - Project Management: Oversaw project timelines and budgets, delivering on time and within budget while mitigating potential risks and roadblocks - Technical Issue Resolution: Led troubleshooting efforts to quickly and effectively resolve platform issues in collaboration with the team <i>Microsystems Engineer</i> March 2022 - December 2022 - Technology Development and Scale-Up: Led the development and automation of scalable microsystem technologies for screening platform, including procurement, installation, and commissioning - Lithography, 3D Printing, and Software Integration: Established lithography and 3D printing facility with comprehensive SOPs, and collaborated with software and data analysis teams deploy multiple software modules - Client Support and Training: Installed screening platforms at client sites and delivered on-site training, ensuring successful implementation and user adoption	

- Business Development: Managed client projects, and crafted business plans focusing on market opportunities, competition analysis, and customer acquisition
- Scientific Representation: Represented NemaLife at scientific conferences, enhancing the company's presence in the scientific community.

Texas Tech University, Lubbock, Texas, USA

Research Assistant

January 2017 - February 2022

- Microfluidics and Automated Systems Development: Completed three microfluidics projects and multiple automated systems, progressing from concept to journal publication
- High-Throughput Microfluidics System: Developed a microfluidics system for whole-life studies in nematodes, improving assay execution time by over 90%, significantly enhancing efficiency compared to traditional methods
- Automated Image Processing Software: Designed MATLAB-based software to process over 6 million images, analyzing nematode phenotype
- Space Biology Collaboration: Collaborated with NASA Space Biology for testing, validation, hardware development, integrating payloads for deployment for muscle strength measurements of *C. elegans* in space aboard the International Space Station (ISS)
- Space Research with ESA and NASA: Worked with NASA and the European Space Agency to study aging-like physiological decline in space flown *C. elegans* treated with resveratrol on the ISS
- Microfabrication and PCB Design: Experienced in lithography-based microfabrication, and proficient in PCB modeling, design, and integration using Eagle and LTspice
- Patent and Startup Formation: Filed a patent for an automated microfluidics system developed during Ph.D. research, which also led to the formation of the startup NemaLife Inc. in 2018

Texas Tech University, Lubbock, Texas, USA

Teaching Assistant

January 2017 - May 2017

- Developed and implemented comprehensive course plans for the semester, teaching senior-level students in Project Lab IV. Delivered engaging lectures, facilitated discussions, and guided students through complex concepts, ensuring academic success and skill development
- Reinforce lessons to students by reviewing material with students in class, one-on-one or in small groups
- Prepare exam materials, quizzes along with tracking attendance and calculating grades
- Provide students feedback on class projects both in theoretical and technical methods.

PUBLICATIONS

Soni, Purushottam; Edwards, Hunter; Anupom, Taslim; Rahman, Mizanur; Lesanpezheshki, Leila; Blawdziewicz, Jerzy; Cope, Henry; Gharahdaghi, Nima; Scott, Daniel; Toh, Li Shean. "Spaceflight induces strength decline in *Caenorhabditis elegans*." Cells, vol. 12, no. 20, 2023, p. 2470. MDPI.

Vintila, Adriana Raluca; Slade, Luke; Cooke, Michael; Willis, Craig RG; Torregrossa, Roberta; Rahman, Mizanur; Anupom, Taslim; Vanapalli, Siva A; Gaffney, Christopher J; Gharahdaghi, Nima. "Mitochondrial sulfide promotes lifespan and health span through distinct mechanisms in developing versus adult treated *CCaenorhabditis elegans*." Proceedings of the National Academy of Sciences, Vol. 120, No. 32, 2023, e2216141120. National Academy of Sciences

Anupom, Taslim; Vanapalli, Siva A. "A compact imaging platform for conducting *C. elegans* phenotypic assays on Earth and in spaceflight." Life, Vol. 13, no. 1, 2023, p. 200. MDPI.

Soni, Purushottam; Anupom, Taslim; Lesanpezheshki, Leila; Rahman, Mizanur; Hewitt, Jennifer E; Vellone, Matthew; Stodieck, Louis; Blawdziewicz, Jerzy; Szewczyk,

Nathaniel J; Vanapalli, Siva A. “Microfluidics-integrated spaceflight hardware for measuring muscle strength of *Caenorhabditis elegans* on the International Space Station.” npj Microgravity, vol. 8, no. 1, 2022, p. 50. Nature Publishing Group

Ellwood, Rebecca A; Hewitt, Jennifer E; Torregrossa, Roberta; Philp, Ashleigh M; Hardee, Justin P; Hughes, Samantha; van de Klashorst, David; Gharahdaghi, Nima; Anupom, Taslim; Slade, Luke. “Mitochondrial hydrogen sulfide supplementation improves health in the *C. elegans* Duchenne muscular dystrophy model.” Proceedings of the National Academy of Sciences, vol. 118, no. 9, 2021, e2018342118. National Academy of Sciences

Zonari, Alessandra; Brace, Lear E; Al-Katib, Kallie Z; Porto, William F; Foyt, Daniel; Guiang, Mylieneth; Cruz, Edgar Andres Ochoa; Marshall, Bailey; Salgueiro, Willian G; Inan, Mehmet Dincer. “Senotherapeutic peptide reduces skin biological age and improves skin health markers.” bioRxiv, 2020, doi:10.1101/2020.10.30.362822. Cold Spring Harbor Laboratory

PATENTS

A Rapid Microfluidic in-vivo Bioassay for Screening Functional Ingredients for Improving Gut Health, 2023 (pending)

Automated Microfluidic System for Lifespan and Healthspan Analysis in Nematodes us 62814110, United States Patent and Trademark Office, 2019 (pending)